Mining the Premier League: Unearthing the Patterns that Lead to Title Glory

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Abstract

In the current world, sports produce considerable statistical information about each player, team, games, and seasons. Traditional sports science believed science to be owned by experts, coaches, team managers, and analyzers. However, sports organizations have recently realized the abundant science available in their data and sought to take advantage of that science through the use of data mining techniques. Sports data mining assists coaches and managers in result prediction, player performance assessment, player injury prediction, sports talent identification, and game strategy evaluation. In this paper, we apply data mining techniques to a decade of English Premier League football data.

1 Introduction

Data mining aims to discover implicit, previously unknown and potentially useful information or knowledge from data. These data can be ranged from traditional shopping cart market data to the emerging social network data. The sheer amount of data available in the world right now has led to an increase in applications of data mining. One of these is the mining of sports data that helps predict future game results for some sports (e.g., football, American football, basketball, hockey) by using historical game results. Many existing sports data mining research papers focus on scheduling of games, visualization of games or players, sport advising, as well as identification and extraction of interesting moments or players from sports game video. While these research papers are interesting and practical, it is also interesting to predict the outcomes of games.

In general, making predictions is not an easy task. Traditionally, some human domain experts (e.g., sports commentators, television and other media personalities, former players, coaches) make predictions on the game results based on their experience, instinct, and/or gut feeling. For example, pre-game analysis of televised sporting events often includes expert predictions. These predictions vary in accuracy as they are typically based on subjective claims and anecdotal evidence. To reduce the subjectivity of making predications based on instinct or gut feeling, a quantitative prediction, which uses some form of objective data to predict the outcome of a game, is used. For example, some professional domain experts make their objective predictions based on historical results. Nowadays, there is a growing contingent in the sports community that seeks to look at this research problem of predicting sports outcomes more analytically. In recent years, simple statistical algorithms (e.g., ranking algorithms) have been applied to historical results in the decision-making process for the prediction of results for future games. These algorithms typically consider the two teams competing and compare their respective strengths and weaknesses in order to make a prediction. Usually, the longer the historical data, the more accurate are the results. However, predictions based on these simple statistical algorithms may not be very accurate when the two teams have not competed with each other.

In this paper, we take the first step in applying statistical analysis and data mining techniques to football data, with a focus on the Premier League, in an effort to make meaningful predictions from data and help validate our own hypotheses about the game.

2 Related Work

The existing potential to analyze and understand large datasets is far below that to collect and maintain data. Therefore, a new generation of techniques and instruments are being developed to aid humans with intelligent

analysis of this high volume of data to produce critical knowledge [1]. Sports provide large amounts of data about each player, team, game, and season and are thus perfect for testing data mining tools and techniques [182 Since experts and statisticians cannot explain relations within data for a single game, data mining techniques are employed to assist the experts or to be used independently in decision making [3]. Consequently, sports teams can gain advantage over their rivals by converting data to applied knowledge through appropriate data extraction and interpretation [4]. Moreover, although various unidentified factors may influence the results, data mining will still be valuable in result prediction [1]. However, it has not been fully exploited in this field. Considering recent research on the use of several data mining techniques including artificial neural networks [5,6,7,8,9,2], decision trees [7,10], Bayesian method [2,11,10,12], logistic regression [2, 10,12], support vector machines [2], and fuzzy methods [7] in predicting sports results, we employ some of these techniques in our work.

2 The Premier League

The Premier League is an English professional league for men's association football clubs. At the top of the English football league system, it is the country's primary football competition. Contested by 20 clubs, it operates on a system of promotion and relegation with the Football League. The Premier League is a corporation in which the 20 member clubs act as shareholders. Seasons run from August to May. Teams play 38 matches each (playing each team in the league twice, home and away), totalling 380 matches in the season. Most games are played on Saturday and Sunday afternoons; others during weekday evenings.

The Premier League is the most-watched football league in the world, broadcast in 212 territories to 643 million homes and a potential TV audience of 4.7 billion people. In the 2014–15 season, the average Premier League match attendance exceeded 36,000, second highest of any professional football league. Most stadium occupancies are near capacity. The Premier League rank second in the UEFA coefficients of leagues based on performances in European competitions over the past five seasons.

While 48 clubs have competed since the inception of the Premier League in 1992, only six have won the title: Manchester United, Chelsea, Arsenal, Manchester City, Blackburn Rovers and Leicester City. The current champions are Leicester City, who won the title in 2015–16.

3 Datasets

We employ two datasets for this paper, using them to not only make predictions over classes, but also to learn association rules and automatically select the best attributes for the job. Since football data is not generally available in the appropriate amount and format for use in data mining tasks, we had to manually build such a dataset in the first place.

3.1 Match Statistics

We obtained match statistics for every game played from the 2005–2006 to the 2014–2015 season, a full decade of football, totalling 3,796 instances. The 36 attributes included contain not only the match statistics but also betting odds for the game. An example instance is given below:

D,D,3,13,2,6,14,16,7,8,0,2,0,0,2.3,3.25,3,2.1,3.25,3.15,2.1,3,3.1,2.1,3,3.2,2.2,3.2,3.2,2.2,3.25,3.1

3.2 Player Starting Positions

The second dataset contains play-by-play statistics of each player of each club for every game of the 2011–2012 season, which comes to a total of 10,369 instances. This is an incredibly detailed dataset with 197 attributes, but it also has the shortfall of being very sparse. We summarize the data by summing up each player's statistics over the entire season, which yields a total of 534 instances. An example instance is given below:

M,25,1466,14,11,3,2,0,1,10,5,4,0,0,0,0,0,0,2,1,2,2,8,3,0,0,2,2,4,0,1,0,0,2,9,5,4,0,0,0,0,0,0,0,0,1,1,0,2,0,0,0,0,0,7,1,0,2,0,0,4,0,0,1,0,0,0,835,273,8,37,811,200,235,28,576,172,68,4,421,64,322,132,751,160,60,40,5,7,24,73,53,1,9,24,18,5,0,22,11,16,43,36,11,5,7,7,29,3,12,4,17,3,12,53,1,7,22,66,13,42,14,48,1321,1,0,1,0,2,5,2,0,0,7,67,103,4,8,63,95,18,10,0,10,8,2,0,1,30,162,20,20,13,3,10,0,0,0,1,3,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,2,0,22,0,0,0,0,42,3,0,368,110,387,243,25,13,3,386,44,2

A complete description of both datasets can be found in the Appendix.

4 Experimental Setup

All experiments were performed in Weka (Waikato Environment for Knowledge Analysis), a popular suite of machine learning software written in Java, developed at the University of Waikato, New Zealand. It is free software licensed under the GNU General Public License.

When learning a classifier for predictions, we use ten-fold cross-validation to track the model's performance on unseen data. This helps ensure that the model makes use of all the data available to it and thus generalizes well.

5 Experiments and Results

We make three different types of investigations into the datasets: 1) predicting match outcomes, 2) selecting the most informative attributes, 3) learning association rules between half-time and full-time results, and 4) predicting players' starting positions.

5.1 Predicting Match Outcomes

We use the first dataset to predict match outcomes as one of three classes: home (H), away (A), and draw (D). For this purpose, we test a number of algorithms and log their respective performances against the baseline performance of a random classifier and ZeroR. Their results are summarized in table 1 on the right. From our experiments, it is clear that the **Logistic** classifier performs best, a simple algorithm that does well to model the few attributes for this elementary classification task. Its confusion matrix is visualized in table 2.

Table 1 Comparison of Algorithms for Match Outcome Prediction

Algorithm	**Accuracy (%)**
Random	33
ZeroR	46.9178
OneR	52.9505
Naive Bayes	45.2845
Decision Table	54.8999
Random Forrest	55.5321
Logistic Regression	59.0358

Table 2 Confusion Matrix for Match Outcome Prediction

		Predicted		
		Away	**Draw**	**Home**
True	**Away**	670	333	62
	Draw	387	424	139
	Home	407	749	625

It is interesting to note the changes in performance caused by only training the classifier on a subset of the attributes. Where a classifier trained on both the match statistics and betting odds achieves an accuracy of 59 %, a classifier trained on only match statistics achieves an accuracy of 55 %, while that trained on only the betting odds achieves an accuracy of 54 %. This shows that betting odds, which are issued well in advance of the match itself, are almost as accurate in predicting the outcome as the real-time match statistics.

5.2 Most Informative Attributes

Although in this dataset we have the luxury of a large number of attributes, it is also interesting to evaluate which attributes are most telling of the final result. For this we use the CfsSubsetEval attribute evaluator with the BestFirst search method. The most informative attributes yielded by this process are listed below:

Match Statistics: AS, HST, AST, HR, AR

Betting Odds: IWH, IWA, LBH, LBA, SBA

These results validate our own hypothesis about the game, since shots and shots on target are important metrics in estimating the outcome of a match. Red cards received by teams are also incredibly influential since the absence of an opposition player makes goal scoring much easier.

Among the betting odds, the most accurate appear to be those issued by Interwetten, Ladbrokes, and Sportingbet. This is in contrast to their popularity among people, since Bet365, Bet&Win, and Gamebookers are used more.

5.3 Learning Relations Between Half-Time and Full-Time Results

In this section, we investigate how the half-time result (HTR) predicts the full-time result (FTR) by mining association rules between the two attributes. We do this via the Apriori algorithm with a minimum confidence of 0.1 in order to accommodate the maximum number of rules. The confidence for each association rule is shown in table 3 below. It is easy to see that the highest confidence is for a home win at half-time to translate into a home win at full-time as well. The next highest is for a half-time away win to translate into a full-time away win. Below that, the confidence of a draw at half-time to remain a draw at full-time as well is quite low, which means that the second half is often used to sway the result in one direction or another. Which team's favour it works in is easily seen by the last two entries, where the home team has a higher chance of turning the result in their favour than the away side.

Table 3 Confidence of Associate Rules

Associate Rule	Confidence
HTR=H → FTR=H	0.82
HTR=A → FTR=A	0.70
HTR=D → FTR=D	0.38
HTR=D → FTR=H	0.37
HTR=D → FTR=A	0.25

5.4 Predicting Player Positions

In this section, we attempt to guess a player's starting position using only details of what he does on pitch. For this task, we use the second dataset that includes play-by-play statistics for each player. We predict the player's position from one of four classes: attack (A), Midfielder (M), Defender (D), Goalkeeper (G). For this purpose, we test a number of algorithms and log their respective performances against the baseline performance of a random classifier and ZeroR. Their results are summarized in table 4 below. From our experiments, it is clear that a Random Forest performs best, which seems logical since it is a classic example of an ensemble that makes use of a large number of simple trees in order to model a complex distribution. Its confusion matrix is visualized in table 5.

Table 4 Comparison of Algorithms for Player Position Prediction

Algorithm	Accuracy
Random	25 %
ZeroR	39.1386 %
OneR	58.427 %
Naive Bayes	64.794 %
Decision Table	69.8502 %
Random Forrest	74.1573 %
Logistic Regression	89.5131 %

Table 5 Confusion Matrix for Player Position Prediction

		Prediction			
		Def	**For**	**GK**	**Mid**
True	**Def**	157	2	0	15
	For	0	95	0	16
	GK	0	0	40	0
	Mid	13	10	0	186

6 Conclusion

In this paper, we set out to apply data mining techniques to unearth patterns and use them to make predictions in different ways. We used a host of different algorithms for each of the two classification tasks, as well as other methods for mining association rules and selecting the best attributes. The Logistic classifier performed best for predicting the outcomes of matches, while a Random Forest achieved the highest accuracy at predicting a player's position on field. To mine association rules, we used the Apriori algorithm, and for attribute selection we employed CfsSubsetEval with BestFirst search. Our experiments yielded encouraging results and ratified many of our hypotheses about the game we have all come to love.

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Appendix A

In this appendix, we describe in detail the datasets used in this paper.

A.1 Match Statistics and Betting Odds

Attributes: 36
Instances: 3,796

FTHG = Full Time Home Team Goals
FTAG = Full Time Away Team Goals
FTR = Full Time Result (H=Home Win, D=Draw, A=Away Win)
HTHG = Half Time Home Team Goals
HTAG = Half Time Away Team Goals
HTR = Half Time Result (H=Home Win, D=Draw, A=Away Win)

HS = Home Team Shots
AS = Away Team Shots
HST = Home Team Shots on Target
AST = Away Team Shots on Target
HC = Home Team Corners
AC = Away Team Corners
HF = Home Team Fouls Committed
AF = Away Team Fouls Committed
HY = Home Team Yellow Cards
AY = Away Team Yellow Cards
HR = Home Team Red Cards
AR = Away Team Red Cards

B365H = Bet365 home win odds
B365D = Bet365 draw odds
B365A = Bet365 away win odds
BWH = Bet&Win home win odds
BWD = Bet&Win draw odds
BWA = Bet&Win away win odds
IWH = Interwetten home win odds
IWD = Interwetten draw odds
IWA = Interwetten away win odds
LBH = Ladbrokes home win odds
LBD = Ladbrokes draw odds
LBA = Ladbrokes away win odds
SBH = Sportingbet home win odds
SBD = Sportingbet draw odds
SBA = Sportingbet away win odds
VCH = VC Bet home win odds
VCD = VC Bet draw odds
VCA = VC Bet away win odds

A.2 Player Statistics

Attributes: 197
Instances: 534

Position Id, Appearances, Time Played, Starts, Substitute On, Substitute Off, Goals, First Goal, Winning Goal, Shots On Target inc goals, Shots Off Target inc woodwork, Blocked Shots, Penalties Taken, Penalty Goals, Penalties Saved, Penalties Off Target, Penalties Not Scored, Direct Free-kick Goals, Direct Free-kick On Target, Direct Free-kick Off Target, Blocked Direct Free-kick, Goals from Inside Box, Shots On from Inside Box, Shots Off from Inside Box, Blocked Shots from Inside Box, Goals from Outside Box, Shots On Target Outside Box, Shots Off Target Outside Box, Blocked Shots Outside Box, Headed Goals, Headed Shots On Target, Headed Shots Off Target, Headed Blocked Shots, Left Foot Goals, Left Foot Shots On Target, Left Foot Shots Off Target, Left Foot Blocked Shots, Right Foot Goals, Right Foot Shots On Target, Right Foot Shots Off Target, Right Foot Blocked Shots, Other Goals, Other Shots On Target, Other Shots Off Target, Other Blocked Shots, Shots Cleared off Line, Shots Cleared off Line Inside Area, Shots Cleared off Line Outside Area, Goals Open Play, Goals from Corners, Goals from Throws, Goals from Direct Free Kick, Goals from Set Play, Goals from penalties, Attempts Open Play on target, Attempts from Corners on target, Attempts from Throws on target, Attempts from Direct Free Kick on target, Attempts from Set Play on target, Attempts from Penalties on target, Attempts Open Play off target, Attempts from Corners off target, Attempts from Throws off target, Attempts from Direct Free Kick off target, Attempts from Set Play off target, Attempts from Penalties off target, Goals as a substitute, Total Successful Passes All, Total Unsuccessful Passes All, Assists, Key Passes, Total Successful Passes Excl Crosses Corners, Total Unsuccessful Passes Excl Crosses Corners, Successful Passes Own Half, Unsuccessful Passes Own Half, Successful Passes Opposition Half, Unsuccessful Passes Opposition Half, Successful Passes Defensive third, Unsuccessful Passes Defensive third, Successful Passes Middle third, Unsuccessful Passes Middle third, Successful Passes Final third, Unsuccessful Passes Final third, Successful Short Passes, Unsuccessful Short

Passes, Successful Long Passes, Unsuccessful Long Passes, Successful Flick-Ons, Unsuccessful Flick-Ons, Successful Crosses Corners, Unsuccessful Crosses Corners, Corners Taken incl short corners, Corners Conceded, Successful Corners into Box, Unsuccessful Corners into Box, Short Corners, Throw Ins to Own Player, Throw Ins to Opposition Player, Successful Dribbles, Unsuccessful Dribbles, Successful Crosses Corners Left, Unsuccessful Crosses Corners Left, Successful Crosses Left, Unsuccessful Crosses Left, Successful Corners Left, Unsuccessful Corners Left, Successful Crosses Corners Right, Unsuccessful Crosses Corners Right, Successful Crosses Right, Unsuccessful Crosses Right, Successful Corners Right, Unsuccessful Corners Right, Successful Long Balls, Unsuccessful Long Balls, Successful Lay-Offs, Unsuccessful Lay-Offs, Through Ball, Successful Crosses Corners in the air, Unsuccessful Crosses Corners in the air, Successful crosses in the air, Unsuccessful crosses in the air, Successful open play crosses, Unsuccessful open play crosses, Touches, Goal Assist Corner, Goal Assist Free Kick, Goal Assist Throw In, Goal Assist Goal Kick, Goal Assist Set Piece, Key Corner, Key Free Kick, Key Throw In, Key Goal Kick, Key Set Pieces, Duels won, Duels lost, Aerial Duels won, Aerial Duels lost, Ground Duels won, Ground Duels lost, Tackles Won, Tackles Lost, Last Man Tackle, Total Clearances, Headed Clearances, Other Clearances, Clearances Off the Line, Blocks, Interceptions, Recoveries, Total Fouls Conceded, Fouls Conceded exc handballs pens, Total Fouls Won, Fouls Won in Danger Area inc pens, Fouls Won not in danger area, Foul Won Penalty, Handballs Conceded, Penalties Conceded, Offsides, Yellow Cards, Red Cards, Goals Conceded, Goals Conceded Inside Box, Goals Conceded Outside Box, Saves Made, Saves Made from Inside Box, Saves Made from Outside Box, Saves from Penalty, Catches, Punches, Drops, Crosses not Claimed, GK Distribution, GK Successful Distribution, GK Unsuccessful Distribution, Clean Sheets, Team Clean sheet, Error leading to Goal, Error leading to Attempt, Challenge Lost, Shots On Conceded, Shots On Conceded Inside Box, Shots On Conceded Outside Box, Team Formation, Position in Formation, Turnovers, Dispossessed, Big Chances, Big Chances Faced, Pass Forward, Pass Backward, Pass Left, Pass Right, Unsuccessful Ball Touch, Successful Ball Touch, Take-Ons Overrun, CompId, SeasId, Touches open play final third, Touches open play opp box, Touches open play opp six yards.